

Context of the Problem

Over the past two years, the growth of **artificial intelligence** (AI) and large scale data centers has greatly increased electricity demand across the United States. In New England (NE), this trend is particularly significant due to emerging “AI campuses” near Boston and Cambridge. According to the Union of Concerned Scientists, major tech companies are starting to explore on-site generation and have even looked into reviving retired nuclear and fossil stations to meet their power needs¹. At the same time, some of New England’s largest power plants (Mystic Generating Station² and the Pilgrim Nuclear Power Station³) have been decommissioned, leaving their existing transmission lines unused yet still valuable.

The emergence of increasing AI-driven load and decreasing energy generation poses a significant challenge for the New England ISO. The region must accommodate substantial demand without causing excessive congestion, reliability risks, or cost escalations. Since new transmissions often require long permitting timelines, **repurposing existing sites** and strategically **adding new resources** at electrically important buses may offer **faster, lower-cost alternatives**. Understanding how these changes affect **emission, pricing, and system flow** is therefore crucial for future planning.

Problem Statement

This project investigates how the New England power system can meet projected AI load growth while managing operational cost, transmission congestion, and carbon emissions. By using the **IEEE-39** bus network as a simplified representation of the ISO-NE grid, the study models baseline system conditions, applies an AI-driven load forecast to 2035, and evaluates two promising strategies: repowering existing generators and building new generation at strategically selected buses.

The analysis uses DC optimal power flow (**DC OPF**) to quantify how each scenario affects locational marginal prices (**LMPs**), line loading, total production cost, and emissions. The goal is to identify which locations and generator upgrades most effectively support the increased load, relieve congestion, and minimize system-wide costs.

Methods and Analysis

1. Load Forecasting via Convex Optimization (Tasks 1 and 2)

- a. **Linear Load Model:** Historical ISO-NE load data for years 2019–2025 were used to estimate the long-term system load trend. Rather than using an unconstrained least-squares regression, the forecast was computed using a convex **quadratic program** (QP) that enforced a **non-negative growth constraint**. Annual average load $L(t)$ was

modeled as a linear function: $L(t) \approx at + b$, $a \geq 0$ in which $X = \begin{bmatrix} t_1 & 1 \\ \vdots & \vdots \\ t_n & 1 \end{bmatrix}$ & $y = \begin{bmatrix} L_1 \\ \vdots \\ L_n \end{bmatrix}$.

The convex optimization problem becomes $\min_{x=[a \ b]^T} \|Xx - y\|^2$ s. t. $a \geq 0$, which expanded gives the QP form: $\min_x \frac{1}{2}x^T Hx + f^T x \rightarrow H = 2X^T X$, $f = -2X^T y$. For my MATLAB implementation, quadprog was used as a form of least-squares QP with linear constraints⁴, in which having $a \geq 0$ stops unrealistic negative-growth scenarios.

- b. **Zonal Forecasting:** Since zone-level historical data was sparse and short-term, a full convex time-series model for each region was not very helpful. Instead, zonal 2035 loads were estimated by a **ratio-based allocation** across the forecast system based on each zone’s **historical average share of total demand**. This approach preserves spatial

¹ <https://blog.ucs.org/mike-jacobs/power-hungry-why-data-centers-are-developing-their-own-energy-sources-to-fuel-ai/>

² <https://www.wbur.org/news/2024/05/30/mystic-electric-station-massachusetts-closing>

³ <https://holtecinternational.com/company/divisions/hdi/our-fleet/pilgrim/>

⁴ <https://www.stat.uchicago.edu/~lekheng/courses/302/notes12.pdf>

proportions observed between 2019–2025 while allowing the system-wide convex model to predict overall growth. Mathematically, this method first computes zone share per year: $s_z(t) = \frac{ZoneLoad_z(t)}{SystemLoad(t)}$, then takes the average across all years: $\bar{s}_z = \text{mean}_t s_z(t)$, and finally scales it to the 2035 system load such that $L_{z,2035} = \bar{s}_z \cdot L_{system,2035}$. This approach assumes each zone consumes roughly the same proportion of total system load every year and that spatial load growth follows the same pattern as historical data.

2. Resource Mix Forecasting and Emissions (Task 2)

- a. **Convex Fit for Resource Shares:** For each generation type g (coal, oil, gas, nuclear, hydro, wind, solar, biomass, other), historical energy shares $\alpha_g(t)$ were fit using a

per-group convex QP: $\min_{a_g, b_g} \sum_i (\alpha_{g,i} - (a_g t_i + b_g))^2$. In this instance, a slope constraint was not needed as some fuel types decrease while others increase. Hence, shares were **normalized** by: $\alpha_g(2035) = \frac{\max(0, a_g t_{2035} + b_g)}{\sum_h \max(0, a_h t_{2035} + b_h)}$, which ensures that the

forecasted mix is non-negative and sums to one.

- b. **Emission Factor Trend:** Using emission coefficients e_g in units of kgCO₂/MWh, the

historical **grid emission factor** becomes: $E(t) = \sum_g \alpha_g(t) e_g$. Another convex QP with non-negative slope was used to fit this data such that: $E(t) \approx \beta(t) + \gamma$, $\beta \geq 0$. The constraint $\beta \geq 0$ prevents the model from predicting a decreasing emissions rate in 2035 if historical data fluctuates, which is conservative because the system will not decarbonize faster until mass structural change occurs.

3. Power System Modeling (Tasks 3–6): All power flow and Optimal Power Flow (OPF) studies used the IEEE-39 bus system⁵ (pictured on the next page) as a reduced AC grid.

- a. **DC Power Flow Approximation:** For all OPF calculations, MATPOWER's DC OPF formulation was used, in which the following **DC power flow assumptions** were made:

Voltage magnitudes are fixed at 1.0 p.u.

Reactive power and voltage constraints are ignored

Small angle approximations s.t. $\sin(\theta) \approx \theta$

Line losses are neglected

Thermal constraints are applied through branch RATE_A.

This gives us the standard DC power flow equation of: $P_{ij} = \frac{\theta_i - \theta_j}{x_{ij}}$ in which P_{ij} is the real power flow and x_{ij} is the line reactance.

⁵ <https://icseg.iti.illinois.edu/ieee-39-bus-system/>

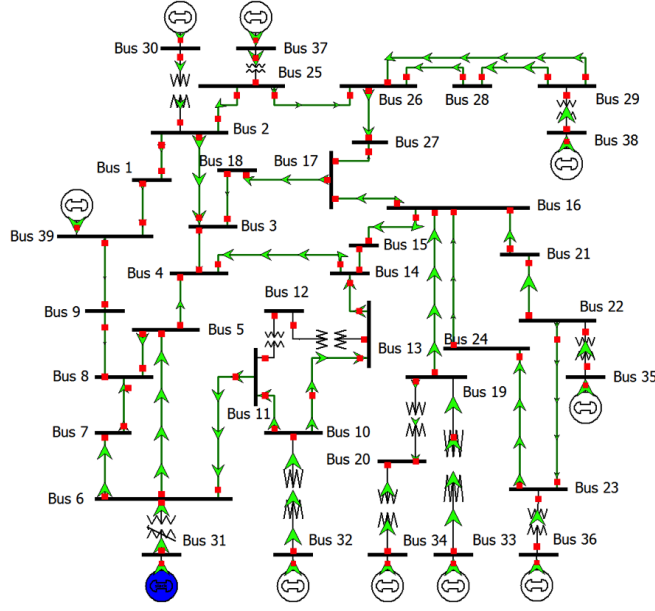


Figure 1: IEEE 39 bus system

b. DC OPF Formulation: DC OPF solves the following: $\min_{P_g, \theta} \sum_g C_g(P_g)$ which is subject

to the following: **Power Balance** $\rightarrow \sum_{g \in G(i)} P_g - P_{D,i} = \sum_j B_{ij}(\theta_i - \theta_j)$,

Generator Limits $\rightarrow P_g^{min} \leq P_g \leq P_g^{max}$, **Line Thermal Limits** $\rightarrow |P_{ij}| \leq P_{ij}^{\text{limit}}$, and

Reference Bus $\rightarrow \theta_{ref} = 0$. The LMP at each bus is solved given spatial price variation

due to congestion: $\lambda_i = \lambda_{sys} + \sum_l \mu_l PTDF_{l,i}$.

4. **Baseline DC OPF (Case 1):** Solved OPF on an unmodified IEEE-39 system, which outputs LMPs, flows, and congestion which will be used as a benchmark for future comparisons.
5. **AI-Driven Load Scenario (Case 2):** Load at buses mapped to high-growth zones is multiplied by the projected **Growth Ratio**, $\frac{L_{2035}}{L_{2025}}$, affecting areas such as Boston, Connecticut, Maine, New Haven, and Rhode Island.
6. **Repowering Existing Generators (Case 3A):** $P_g^{max, new} = 2P_g^{max}$ & $C_g^{new}(P_g) = 0.7C_g(P_g)$.

We **double** $P_g^{max, new}$ as it is consistent with typical repowering outcomes for aging natural-gas and oil units where older turbines (100 to 400 MW) are replaced with modern combined-cycle blocks capable of 600–800 MW at similar sites. Also, repowered units generally **burn fuel more efficiently**, as modern combined cycle gas turbines (CCGTs) heat rate drops lead to drastic improvements⁶. Furthermore, transmission rates are increased 1.2 times: $P_{ij}^{new} = 1.2P_{ij}$ as repowering often requires modest **transmission upgrades** such as new breakers, reconductoring, and cooling improvements.

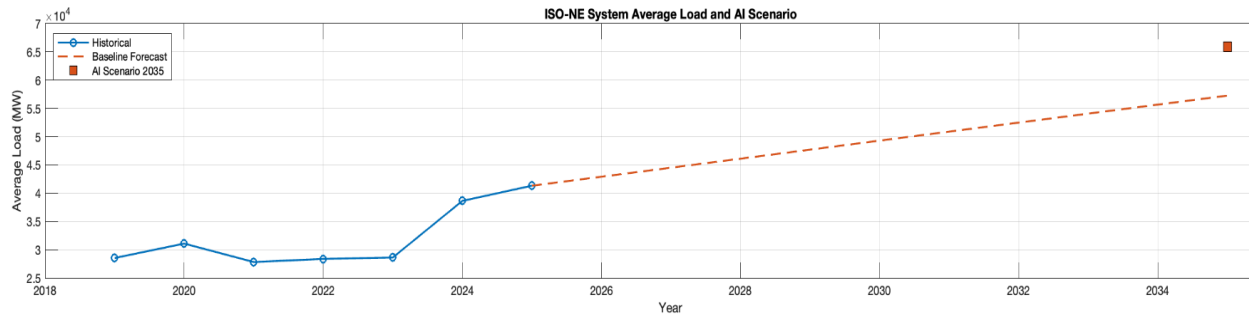
⁶ <https://www.eia.gov/todayinenergy/detail.php?id=60984>

7. **Building New Generation:** New generators are added at buses in high-growth regions that previously had no generation. AI-driven load introduces uncertainty in both peak demand and hourly flexibility. Planning criteria from NERC/ISO standards typically require **reserve margins** of 15 to 20%. Thus, we have a **capacity size** $P_{g,new}^{max} = \frac{1.5P^{needed}}{N_{new\ buses}}$. Also, these new units use a simplified quadratic cost of $C(P) = 20P$ as new gas units in ISO-NE have marginal costs between \$18-\$32/MWh depending on fuel prices, making $20P$ a competitive gas turbine marginal cost. Transmission around new buses were strengthened accordingly: $P_{ij}^{new} = 3P_{ij}$ if the line touches the new generation bus, as new generation interconnections typically require major **transmission reinforcement** at the point of intersection or POI.
8. **Emissions Modeling:** Emissions are accounted for using: $E = \sum_g P_g e_g$ with e_g taken from Task 2's forecasted emission factors. This ties generation changes from Tasks 5 and 6 to **system-wide carbon impact**.

Simulations

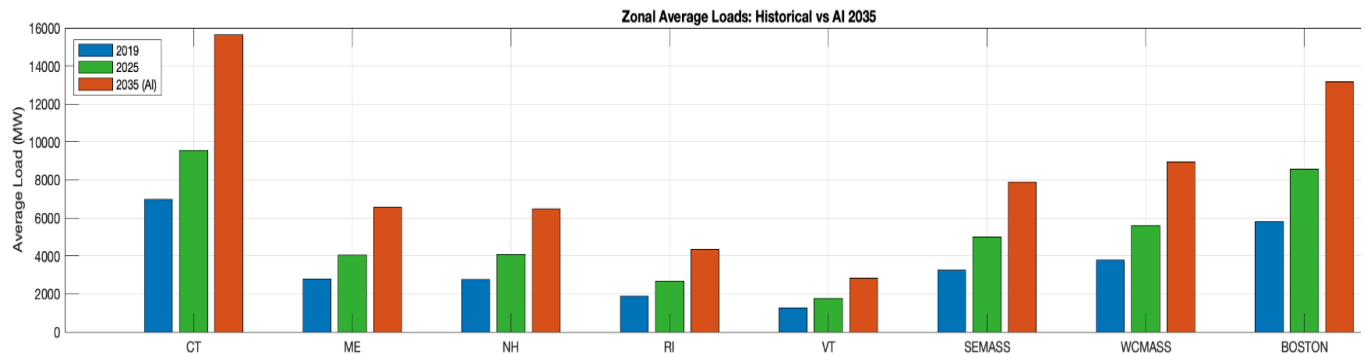
1. **Overview:** All simulations were implemented in MATLAB using **MATPOWER 8.1** for DC OPF and AC power flow checks. The main script sequentially executes the following six tasks: 1) Historical and AI-driven load forecasting for ISO-NE; 2) Resource-mix and emissions forecasting; 3) A 39-bus baseline DC OPF (Case 1); 4) An AI-driven load scenario (Case 2); 5) A repowering scenario with upgraded existing plants (Case 3A); 6) A new-generation build-out scenario (Case 3B). For each case, the code reports system load, total operating cost, line loadings, LMPs, and in the summary, approximate carbon emissions. The figures in the report are generated automatically at the end of the script and saved as PNGs.
2. **Task 1 – ISO-NE Load Forecast under AI Growth:** For Task 1, I first computed historical average system load using ISO-NE three-day reliability region demand forecast data from 2019 to 2025⁷. For each year, the script aggregates all 72-hour forecasts across all files, sums the MW columns, and divides by the total number of hours to obtain an annual average load, which ranges from roughly 27 to 31 GW through 2023, then jumps to 38.7 GW in 2024 and **41.3 GW in 2025**, reflecting recent load growth.

A convex least-squares fit is then applied to the historical points by solving a small quadratic program for slope a and intercept b subject to $a \geq 0$, which enforces a non-decreasing load over time. This line defines the baseline **2035 forecast of 57,277 MW**. To further represent AI-driven data center growth, I applied an extra multiplicative factor of 15%, resulting in an **AI 2035 forecast of 65,868 MW**. The left panel of the figure below shows the historical averages (blue line from 2019 to 2025), the linear baseline projection (orange dashed line from 2026 to 2035), and the AI scenario point (orange box in 2035).



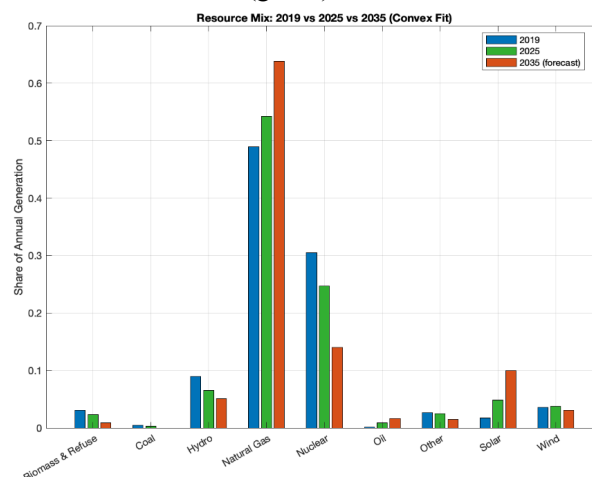
⁷ <https://www.iso-ne.com/isoexpress/web/reports/load-and-demand/-/tree/three-day-reliability-region-demand-forecast>

Zonal shares are computed by averaging the ratio of each zone's load to the total system load across all historical years. These shares are then applied to the AI 2035 system load to obtain **zonal projections**. The resulting map shows that **Connecticut, Boston, Western/Central MA, and SEMASS** account for most of the AI-driven growth, with Connecticut alone exceeding **15.6 GW** and Boston exceeding **13.1 GW** in **2035** (in orange). The chart below shows zonal averages that have been recorded in 2019 (blue) and 2025 (green) and the AI 2035 projection (orange).



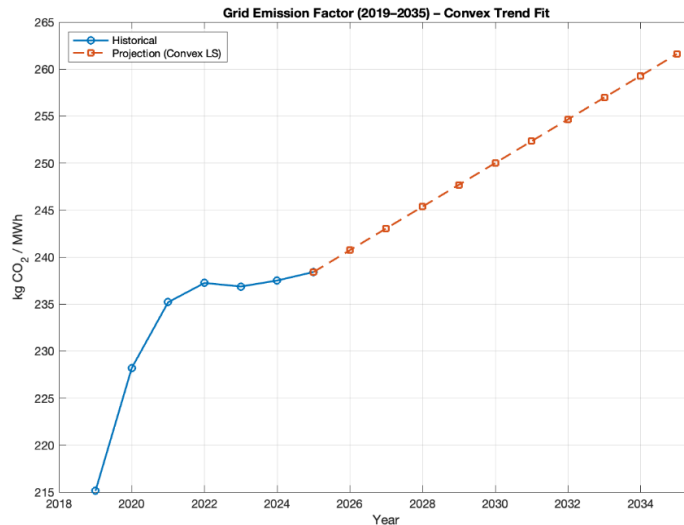
- Task 2 – Resource Mix and Emission Factor Forecast:** Task 2 builds on the load forecast by estimating how New England's generation mix, and its carbon intensity, may change through 2035. Using ISO-NE's annual "energy peak by source" data⁸, monthly energy production values for each resource type are aggregated to obtain yearly totals. These resources are grouped into nine broad fuel categories: coal, oil, natural gas, nuclear, hydro, wind, solar, biomass/refuse, and "other" categories. For each year of historical data, the share of annual energy supplied by each group is computed relative to the total energy produced in that year.

To project the future mix, a **convex trend** is fitted to each fuel group. This type of trend ensures smooth, gradually changing shares rather than sudden jumps, and it avoids unrealistic behavior such as a fuel source rapidly becoming negative or dominating the entire supply. The resulting 2035 projection reflects a continuation of current trajectories: **natural gas remains the largest contributor** to annual energy supply, nuclear and hydro continue to provide stable zero-carbon baseload, and wind and solar slowly increase but still represent a small portion of the mix. The left panel of the Task 2 figure below illustrates this evolution by showing the earliest 2019 mix (blue), the most recent 2025 mix (green), and the forecasted 2035 mix (orange).



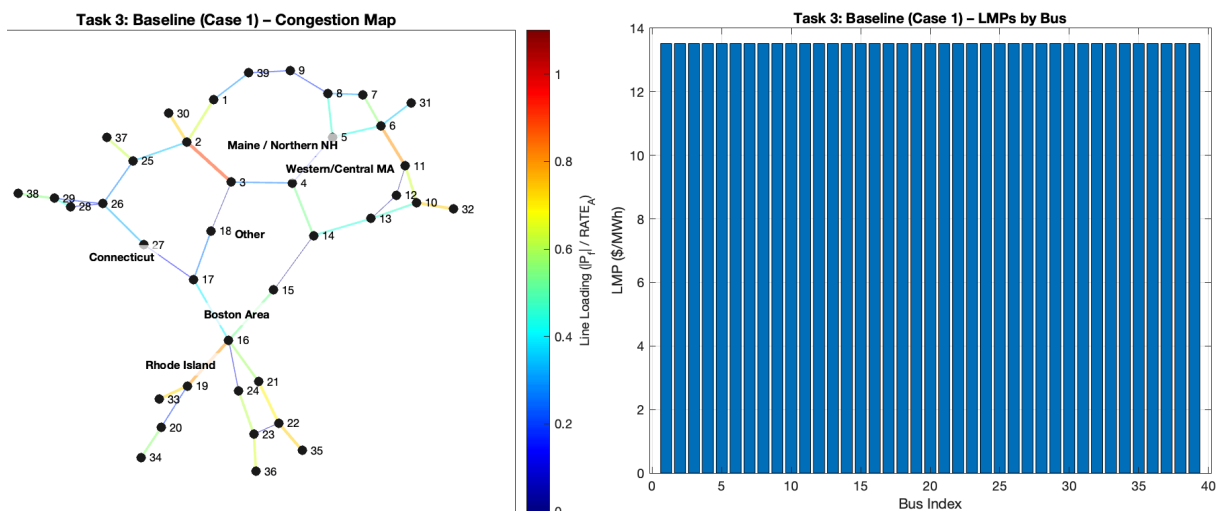
⁸ <https://www.iso-ne.com/isoexpress/web/reports/load-and-demand/-/tree/net-ener-peak-load>

To translate these mix projections into a systemwide carbon intensity, each fuel category is associated with a representative emissions rate (in kilograms of CO₂ per megawatt-hour). Fossil fuels such as coal, oil, and natural gas have relatively high intensities, while nuclear, hydro, wind, and solar are treated as zero-carbon. The overall grid emission factor for a given year is then determined by weighting each fuel's **intensity** by its **share** in that year's resource mix. The historical emission factors form an **increasing** trend, reflecting the **reintroduction of oil and gas-fired generation** and the **slow adoption of renewable growth**.



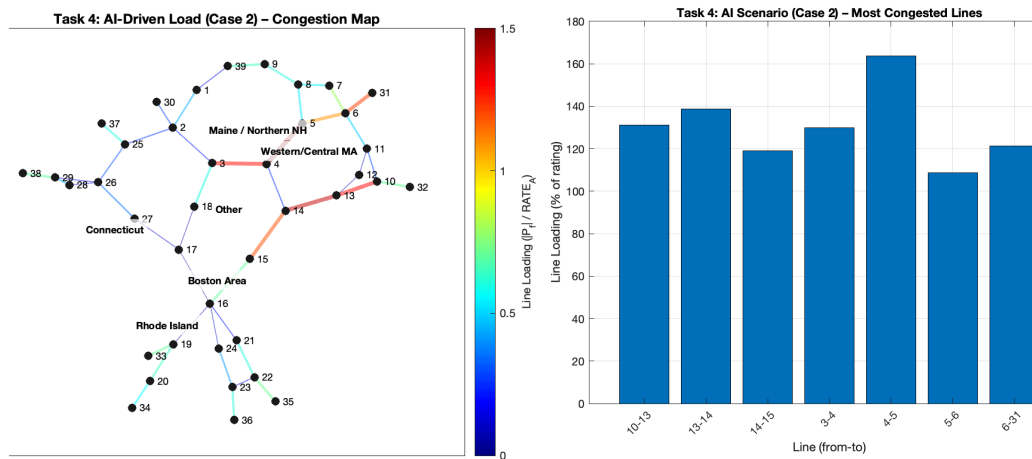
4. **Task 3 – IEEE-39 Baseline DC-OPF (Case 1):** Task 3 establishes a baseline operating point using the standard MATPOWER 39-bus test case. A DC OPF is solved with the historical generator cost coefficients and transmission ratings. The solution is feasible and yields a total load of **6,254.2 MW** and total system cost of **\$41,263.9/h**. Since the network is **uncongested** under this loading, **LMPs are uniform at \$13.52/MWh** across all buses.

The code includes a mapping from buses to approximate ISO-NE regions (Boston, Connecticut, Maine/NH, Western/Central MA, Rhode Island, and “Other”) so that representative, regional LMPs can be made. The congestion map in the figure below shows all lines with loading **well below their thermal limits** (colors close to blue), and the LMP bar chart displays a **flat price profile** across all 39 buses. These baseline outputs provide the reference point for later comparisons of cost, congestion, and emissions.



5. **Task 4 – AI-Driver Load Scenario:** Task 4 applies the AI-driven demand growth from Task 1 directly to the IEEE-39 bus system above. The projected increase in New England’s system-wide load is converted into a single growth factor, which is then applied selectively to the buses that represent **high-growth areas** such as Boston and Connecticut. All other regions retain their baseline loads, reflecting geographically uneven development of **AI campuses** and **data centers**.

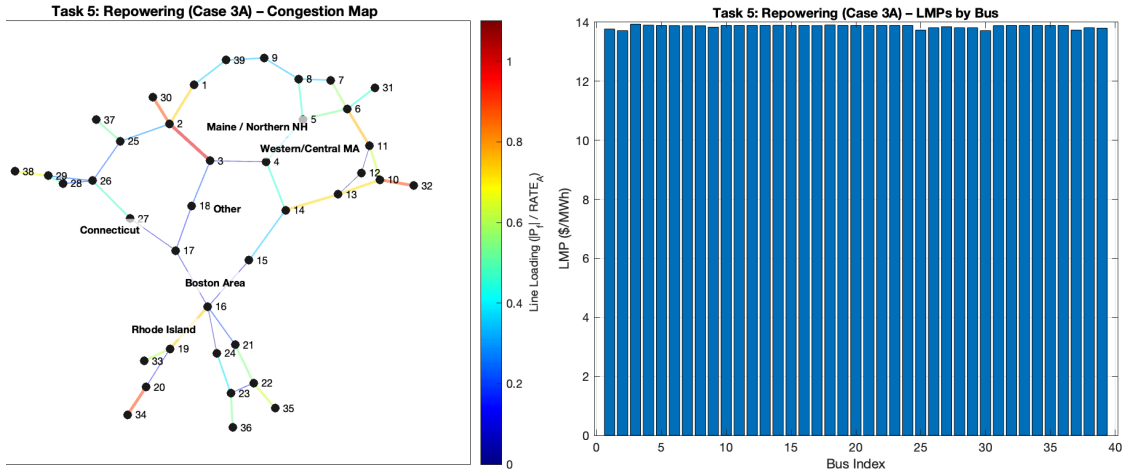
After scaling these regions, the total load in the 39-bus system rises by roughly **one-third** relative to the baseline case! Running a DC OPF on this stressed system reveals immediate consequences: **multiple lines** approach or exceed their **thermal limits**, **congestion** becomes **widespread**, and some buses become heavily overloaded. In several runs, the system is unable to find a feasible dispatch, showing that the existing transmission network and generator fleet **cannot reliably serve future AI-era demand without significant upgrades**. This facilitates the need for repowering and new-generation strategies explored in Tasks 5 and 6.



6. **Task 5 – Repowering Existing Generators (Case 3A):** Task 5 evaluates whether the grid can recover from the infeasibility observed in the AI-driven load case above by strengthening existing generation and transmission assets rather than building new ones (analogous to reopening a decommissioned power plant). The strategy focuses on **repowering generators** located in the regions most affected by AI-driven demand growth (Boston, Connecticut, Maine/NH, and Rhode Island). These sites are natural candidates because they already host large power plants, substations, and high-capacity transmission corridors.

Repowering is modeled by significantly increasing the output capability of these generators and making them more efficient, which represents **real-world upgrades** such as turbine replacements, combined-cycle retrofits, or modern emission-controlled units installed at outdated sites. At the same time, transmission lines across the system receive **moderate capacity increases** due to cost-effective upgrades like reconductoring or improved thermal ratings.

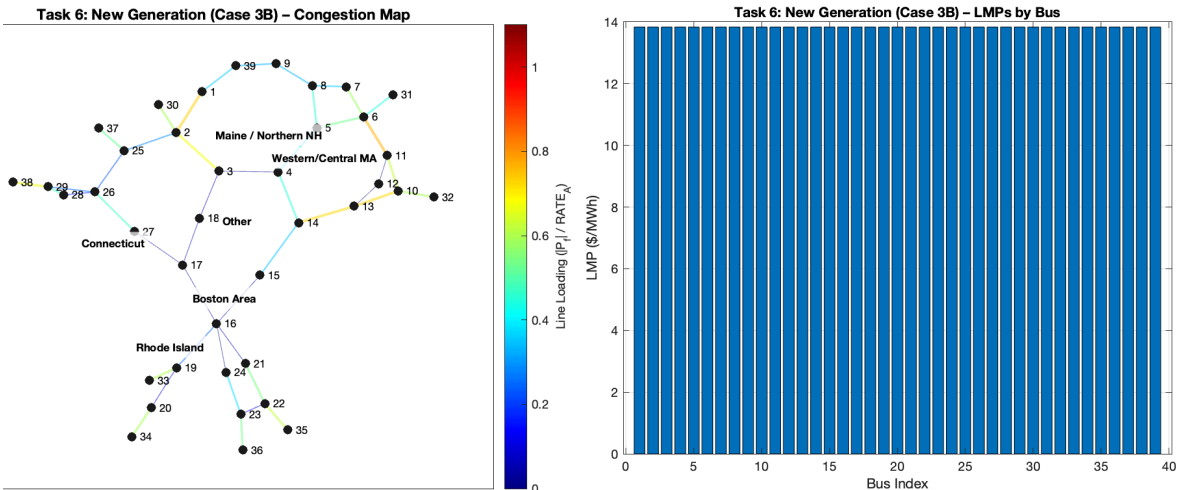
When these reinforcements are applied, the system becomes far more stable under AI-era loads. **Congestion is reduced, dispatch costs fall** compared to the overloaded scenario in Task 4, and the **OPF becomes feasible again!** However, congestion does not disappear entirely, and several buses still operate close to their limits, indicating that repowering alone is not enough to fully resolve the long-term stresses imposed by future data-center growth. This motivates the expansion of new generating capacity explored in Task 6.



7. **Task 6 – New Generation Build-Out (Case 3B):** Task 6 explores a complementary strategy: **adding new generation at previously non-generator buses** in high-growth regions. Starting from the repowered case (Case 3A), the code identifies candidate buses in Boston, Connecticut, Maine/NH, and Rhode Island that currently have **no generation**. It selects up to four of these buses and installs **new gas generators** sized to cover the remaining capacity gap under AI load with a safety margin. In my simulation, four new plants of **187.5 MW** each are added (total nominal new capacity 750 MW) at buses 3, 16, 17, and 18.

Each new generator is modeled with a linear marginal cost of **\$20/MWh** (gas-fired plant) with **no startup/shutdown cost**. Transmission ratings are further increased on lines incident to new generator buses (factor of 3.0) and on lines that were above 80% loading in the repowering case (factor of 1.5). This mimics targeted line uprates alongside new plant construction.

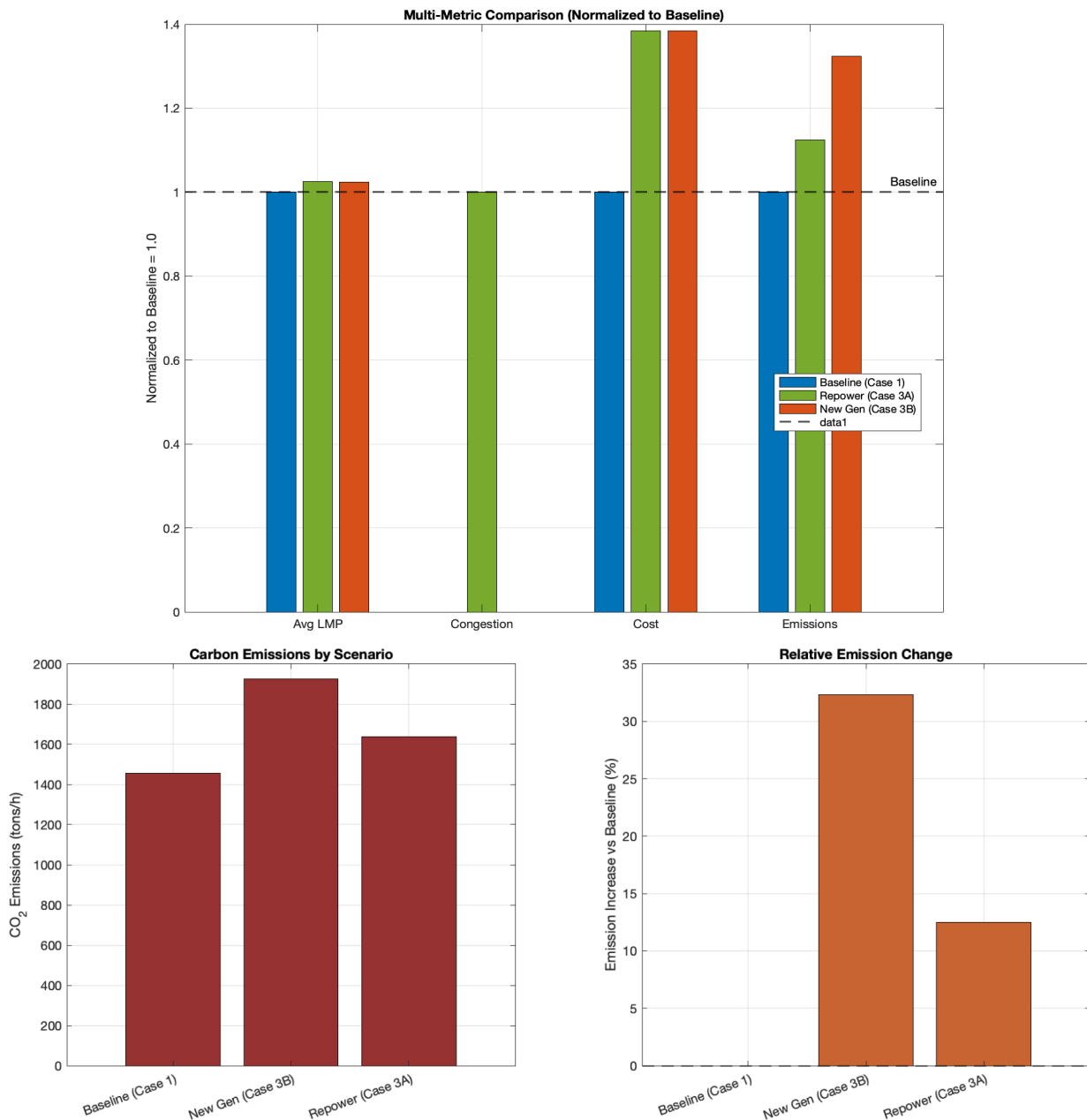
The resulting DC OPF for Case 3B is feasible and produces almost **uncongested operation** as the summary table reports **no lines above the 95% loading threshold!** Total operating cost is **\$57,141.9/h**, slightly lower than in Case 3A, and the LMPs become nearly uniform again, with an average of **\$13.84/MWh**. The congestion map and LMP bar plot for Task 6 confirm that both thermal and price spreads are much smaller than in Case 3A.



8. **Summary of Cost, Congestion, and Emissions:** The summary table shows that:
- Baseline (Case 1): \$41,264/h, 0 congested lines, avg LMP \$13.52/MWh;
 - AI Load (Case 2): infeasible under existing limits;
 - Repower (Case 3A): \$57,143/h, 1 congested line, avg LMP \$13.86/MWh;
 - New Gen (Case 3B): \$57,142/h, 0 congested lines, avg LMP \$13.84/MWh.

Emissions are approximated using a blended emission factor for existing generators (based on Task 2) and a fixed 400 kgCO₂/MWh for new gas units. Baseline emissions are 1,455.1 tons CO₂/h. Under Case 3A, emissions increase to 1,637 tons/h (+12.5% vs baseline), driven by higher load with only modest improvements in efficiency. Case 3B reaches 1,925 tons/h (+32.3% vs baseline), reflecting both higher output and higher emission factor for new gas capacity.

The three summary figures in the report present: (i) average LMPs and LMP distributions by scenario, (ii) absolute and relative emissions, and (iii) a multi-metric comparison where cost, congestion, average LMP, and emissions are all normalized to the baseline of 1.0. Together, these simulations illustrate that while **repowering** and **new generation** can **restore feasibility** and **control congestion** under AI-driven load growth, they come with **higher operating costs** and **significantly higher carbon emissions** unless accompanied by deeper decarbonization of the resource mix.



Tasks and Deliverables

Task 1 – Predict AI-Driven Load Growth – **100%**

Task 2 – Predict Future Resource Mix – **100%**

Task 3 – Build and Validate IEEE-39 Baseline Model/Case 1 – **100%**

Task 4 – Simulate AI-Load Growth Scenario/Case 2 – **100%**

Task 5 – Repower Existing Generators/Case 3A – **100%**

Task 6 – Add New Generation Resources/Case 3B – **100%**

Conclusions

This project explored how the New England power system might respond to rapid AI-driven load growth using a sequence of increasingly aggressive interventions. The results show that **simply absorbing the demand increase** (Task 4) is **infeasible** with today's grid, and that **repowering existing generators** (Task 5) **provides meaningful relief** but still **leaves several corridors stressed**. Adding **new, strategically located generation** (Task 6) further **improves feasibility**, **reduces LMPs**, and **lowers systemwide congestion**, though it comes with **higher carbon emissions** if the new units rely on **natural gas technologies** as is expected by today's trendlines. Together, these outcomes highlight a broader theme: no single measure is sufficient on its own. A realistic transition will likely require a **combination** of repowering legacy sites and targeted deployment of new resources in high-growth areas.

In the future, I would like to integrate Tasks 5 and 6 into a unified **Case 4** that simultaneously upgrades existing generation, expands new capacity, and co-optimizes transmission enhancements. This combined approach would more closely resemble real system-planning practice, where multiple approaches such as **repowering**, **new generation**, and **line upgrades** are evaluated together rather than in isolation. The analysis in this report has helped me understand how load forecasting, convex optimization, DC/AC OPF modeling, and congestion analysis interact in a modern grid-planning workflow. Most importantly, it shows how sensitive power-system feasibility and costs are relative to where demand grows and where new resources are added. As AI-related electricity consumption continues to rise, these types of planning tools will be essential for ensuring New England can maintain **reliability**, **affordability**, and **sustainability**.

Sources/References/Links

All external sources referenced in this report are cited in footnotes on the corresponding pages. **ChatGPT was used** to assist with figure creation in MATLAB and to improve code formatting and readability.

The following are links relevant to this report:

Google Drive: <https://drive.google.com/drive/folders/1GpR9Z-VAhdrCOdUSRzOicRXUxyPUxhZI>

Figures: <https://drive.google.com/drive/folders/195ZYKSjyXe6QBUK6Pks-41cTMOZQRmjE>

Historical CSVs: https://drive.google.com/drive/folders/10HD0joLXqZPZwVABDzwZfit_drYj35Vu

Mix Excel Files: <https://drive.google.com/drive/folders/1rIGu2QbEx5fmfwJwuwwq20XTGLhyacglx>

MATLAB File: https://drive.google.com/file/d/1n3tbrp7eJ_46sguY97qgCTYKka0wQoo